Exercises

Algebraic problems

Question 5.1: Find the second order derivative of the following functions (we have seen two of these formulas when we practised using the product and quotient rule):

(a) $y = \sqrt{x^2 + 25}$
(b) $y = \frac{x^5}{4} + x^3 + 12$
(c) $f(x) = x\exp(x)$
(d) $P(z) = P_0 \exp\left(-\frac{g}{R_a T} z\right)$ which is the same simplified form of the hydrostatic equation we have met before, where P_0, g, R_a and T are constants.

Question 5.2: Find the stationary values of the following functions:

(a) $f(x) = \frac{x^3}{3} + \frac{x^2}{2} - 30x + 5$
(b) $f(x) = x^3$

Verify whether the stationary values are maxima, minima or saddle points.

Question 5.3: Complete Chapter 5 Exercise 3, and use Newtons Method to find the root (accurate to 6 decimal places) of following function:

$$f(x) = 2x^2 + 5 - e^x$$

Assume the root is between $x = 3$ and $x = 4$. Newtons method is used for finding the root, so you need to move all terms to one side. Use a first guess of 3.5.

Question 5.4: Using Newtons method, find the second guess of the root (i.e. x_2) for the following functions:

(a) $f(x) = x^3 - 7x^2 + 8x - 3$, with $x_0 = 5$
(b) $f(x) = sin(x) - ln(x)$, with $x_0 = 2$
(c) $f(x) = xcos(x) - x^2$, with $x_0 = 1$

Question 5.5: Use Newtons method to find all the roots of the following equation to 3 decimal places. You may use a starting guess of $x_0 = -3.5$ and $x_0 = 4.5$ to find the left and right sided roots respectively:

$$x^3 - x^2 - 15x + 1 = 0$$

Take a look at Figure 5.3. We seem to have 3 roots, so use this information to confirm the values of these roots to within 3 decimal places.

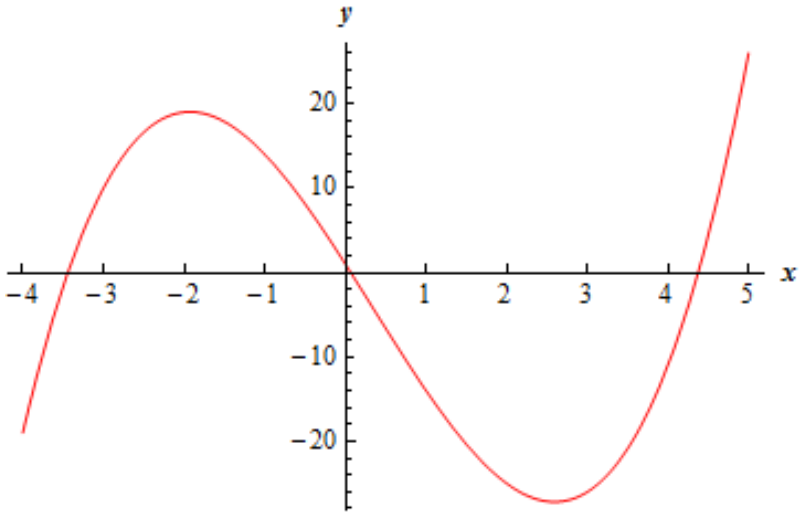


Figure 5.3: Function that varies according to a cubic function

Real-world problems

Question 5.6: An industrial incinerator produces a steady output of particulate matter having a concentration of $C_0 = 1.0$ mg m^{-3} of air at the chimney. 1 km away from the incinerator plant the concentration is $C_{1\,km} = 0.1$ mg m^{-3}. Some houses are a distance of 500 m away from the plant. The equation that lets us calculate the concentration as a function of distance from the incinerator is provided by the following expression.

$$C(x) = C_0 + \frac{C_{1\,km} - C_0}{1000}x$$

Prove that the flux is in steady state.

Question 5.7: At 3km below the surface there is a hot granite bed, with a temperature of 250 °C. The temperature just below the surface the temperature stays relatively constant at 10°C. Fourier's law in 1-d has been used to construct the following expression.

$$T(x) = 10 + \frac{250 - 10}{3000}x$$

Is the system in steady state?

Question 5.8: Imagine you are a business analyst for a company that sells a product. The company wants to model its gross profit, $P(x)$, based on the number of units sold, where x represents the number of units sold in thousands. The profit function is given by:

$$P(x) = e^x - 3x^2$$

(a) Determine the number of units sold where the gross profit is at its minimum. *HINT: You may need to apply Newton's method to find the root of the derivative in order to locate the stationary point. Use a starting guess of $x_0 = 3$ and find the root to 3 decimal places.*

(b) Confirm that the gross profit is minimised at this value.